Brittany Terese Fasy

CONTACT INFORMATION

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EDUCATION

Ph.D., Computer Science, 2012
Duke University, Durham, NC

Thesis: *Modes of Gaussian Mixtures and an Inequality for the Distance Between Curves in Space*
Adviser: Herbert Edelsbrunner

B.S., Mathematics and Computer Science, 2007
Saint Joseph's University, Philadelphia, PA

Honors Program. Summa Cum Laude.
Thesis: *Homotopy Classification of the Components of the Space of Maps into an Aspherical Space: a Problem in the Intersection of Group Theory and Topology*
Adviser: Samuel Smith

RESEARCH POSITIONS

Associate Professor, Gianforte School of Computing, 2021–Present
Affiliate Faculty, Mathematical Sciences, 2017–Present
Gianforte Faculty Fellow, 2015–Present
Assistant Professor, Gianforte School of Computing, 2016–2021
Assistant Professor, Computer Science, 2015–2016 Montana State University, Bozeman, MT

Postdoctoral Researcher and Newcomb Fellow, Computer Science, 2013–2015
Tulane University, New Orleans, LA
Adviser: Carola Wenk

Postdoctoral Fellow, Computer Science, 2012–2013
Carnegie Mellon University, Pittsburgh, PA
Adviser: Gary Miller

Visiting Scientist (Graduate RA), 2009–2012
IST Austria, Vienna, Austria

College Student Senior Technical Specialist, May 2006–August 2008
College Student Technical Specialist, May 2006–May 2007
Lockheed Martin, King of Prussia, PA
Lockheed Martin, Gaithersburg, MD

University Scholar Researcher, Mathematics, 2006–2007
Saint Joseph's University, Philadelphia, PA
Adviser: Samuel Smith.

Summer Research Scholar, Computer Science Education, 2004–2005
Saint Joseph's University, Philadelphia, PA
Adviser: Stephen Cooper.

RESEARCH GRANTS

(current) *CAREER: Topological Descriptors*
NSF CCF 2046730, 1 July 2021–30 June 2027 (est.), $599,348.
PI: BTF.

CDS&E: Extracting Physics from High-Fidelity Simulations of Atomization using Geometric and Topological Data Analysis
NSF CBET 2152737, 1 July 2022–30 June 2026, $375,271
PI: M. Owkes; co-PI: BTF; former co-PI: D. L. Millman.

CS4ALL: Indian Education in Computing: A Montana Story
1 September 2020–28 February 2026, NSF DRL 2031795, $635,255.
PI: BTF; former co-PIs: S. Hancock, T. Peters.
Collaborative Grant: DRL 2031279 (PI Searle, Utah State U.; co-PI Tofel-Grehl).

FRG: Statistical Approaches to Topological Data Analysis that Address Questions in Complex Data
1 September 2019–31 August 2025, NSF DMS 1854336, $404,183.
PI: BTF. CMU Subcontract: A. Rinaldo (PI) and L. Wasserman (former co-PI).
Collaborative Grant: DMS 1854220 (PI J. Cisewski-Kehe, Yale and U. Wisconsin Madison).

ITEST: Improving the Pathway for Rural and American Indian Students Entering CS Via Storytelling
1 August 2017–31 July 2024, National Science Foundation, DRL 1657553, $1,149,691.
PI: BTF; co-PIs: S. Hancock, B. Komlos, S. Windchief, and M. Wittie.
- REU Supplement: Summer 2018 for $16,000.

QuBBD: Quantifying Morphologic Phenotypes in Prostate Cancer - Developing Topological Descriptors for Machine Learning Algorithms
1 August 2017–31 July 2023, co-funded by NIH and NSF, DMS 1664858, $420,706.
PI: BTF; co-PI J. Sheppard. Collaborative Grant: DMS 1557750 (PI C. Wenk, Tulane).

AF: Small: Geometric and Topological Algorithms for Analyzing Road Network Data
1 July 2016–30 June 2020, National Science Foundation, CCF 1618605, $152,824.
PI: BTF. Collaborative Grants: CCF 1618469 (PI C. Wenk, Tulane) and CCF 1618247 (PI Y. Wang, Ohio State).
- REU Supplement: Summer 2017 for $19,000 (funded two REU students).

ABI Innovation: Biofilm Resource and Information Database (BRaID): A Tool to Fuse Diverse Biofilm Data Types
1 August 2017–31 July 2020, National Science Foundation, DBI 1661530, $299,853.
PI: David Millman; co-PI: Brendan Mumey. Collaborative Grant: DBI 1661527 (PI Ramaraj, NCGR).
My Role (1 August 2017–1 May 2019): Consulting on the use of topological descriptors.

Planning Grants

S-STEM: Building a Montana Computing Consortium
NSF DUE 2221684, 1 July 2022–30 June 2026 (est.), $99,836.00.
PI: BTF; co-PIs B. LaMeres, D. L. Millman, B. Zhu.

QuBBD: Towards Automated Quantitative Prostate Cancer Diagnosis
15 September 2015–31 August 2017, co-funded by NIH and NSF, DMS 1557716, $46,639.
PI: BTF. Collaborative Grant: DMS 1557750 (PI C. Wenk, Tulane)

Conference Grants

Algebraic Topology: Methods, Computation, and Science 2025
NSF DMS 2520735, 15 June 2025–31 May 2026, $24,999. PI: BTF; co-PI: F. Mémoli (Rutgers).

Algebraic Topology: Methods, Computation, and Science 2022
NSF DMS 2208855, 1 June 2022–31 May 2024, $26,000.
PI: M. Kahle (Ohio State U.); co-PIs: BTF, R. Ghrist (U. Pennsylvania), S. Weinberger (U. Chicago).

Topology for Data Science: An Introductory Workshop for Undergraduates
NSF DMS 1955925, 15 January 2020–30 June 2023, $30,474.
PI: BTF; Former PI and co-PI: S. Hancock and D. L. Millman.

Workshop for Women in Computational Topology
NSF DMS 1619908, 15 June 2016–31 January 2018, $30,000.
PI: L. Ziegelmeier (Macalester); co-PIs: BTF and E. Chambers (St. Louis U).

- Additional support from Microsoft Research: $5,000.
- Funding for follow-up meetings from IMA: $15,000.

REU/RET Grants (current) *REU Site: Algorithms in Data Science with Applications*
NSF CCF 2548411, 1 May 2026–30 April 2029, $464,998
PI: A. L. Ralalage; co-PI: B. Mumey.
My role: SP (REU mentor).

REU Site: Algorithms and Optimization for Sustainability and Biology.
NSF CNS 2243010, February 2023–January 2026, $323,949.
PI: S. Yaw; co-PI: B. Mumey.
My role: SP (REU mentor).

RET Site: Innovative Transportation Systems
NSF EEC 1610089, October 2017–September 2023, $594,426.
PI: C. Woolard; Co-PIs J. Stephens and S. Gallagher.
My role: SP (RET mentor and workshop facilitator).

REU Site: Research and Development of Algorithms in a Software Factory.
NSF CCF 1658971, March 2017–February 2021, $288,000.
PI: C. Izurieta; co-PI: B. Mumey.
My role: SP (REU mentor).

Postdocs

Alex McCleary, Computational Topology. Aug. 2023–Present.

Jake Chipps, Computer Science Education, co-advised with Stacey Hancock. Jan. 2020–June 2023. Now the STEM Education Researcher-Practitioner at the American Computer and Robotics Museum in Bozeman, MT.

PhD Advisees

Madhav Cherupilil Sajeev, current Ph.D. student in Computer Science.

Paul Cornish, current Ph.D. student in Computer Science.

Jacob Sriraman, current Ph.D. student in Computer Science.

Bradley McCoy, Ph.D. in Computer Science, MSU (May 2024). Dissertation entitled "From Curves to Words and Back Again: Geometric Computation of Minimum-Area Homotopy." Now a tenure-track Assistant Professor at James Madison University.

Anna Schenfisch, Ph.D. in Mathematics, MSU (May 2023). Dissertation entitled "Faithful Sets of Topological Descriptors and Algebraic K-Theory of Multi-Parameter Zig-Zag Persistence Modules." Now a postdoc at KTH Royal Institute of Technology.

Sushovan Majhi, co-advised with C. Wenk, Ph.D. in Mathematics from Tulane U. (Dec. 2020). Dissertation entitled *Topological Methods in Shape Reconstruction and Comparison.* Now an non-tenure-track Assistant Professor of Data Science at George Washington University.

Samuel Micka, Ph.D. in Computer Science from MSU (May 2020). Dissertation entitled *Searching and Reconstruction: Algorithms with Topological Descriptors.* Now an Assistant Professor of Computer Science at Western Washington University.

MS Advisees (Thesis)

Olivia Firth, M.S. in CS, MSU (August 2022). Thesis entitled *Surveying Middle School Computer Science Throughout the United States: A Needs Assessment for Teachers.* Now Program Coordinator in the MSU Science Math Resource Center.

Graduate RAs

Eli Quist, M.S. in CS, MSU (RA in 2025).

Jack Ruder, M.S. in CS, MSU (RA in AY 2024–2025).

Braeden Sopp, M.S. student, Computer Science, MSU (RA 2022–2024).

Nathan Patera, M.S. in CS, MSU (December 2023).

Elijah Meyer, Ph.D. in Statistics, MSU (RA Spring 2022).

Therese Luapriello, M.S. in Statistics, MSU (RA 2021–2022).

James Wilson, M.S. in Data Science, MSU (RA 2020–2021).

Jordan Schupbach, Ph.D. in Statistics, MSU (partially or fully RA 2017–2021).

Alex Mershon, M.S. in CS, MSU (RA Spring 2020).

Monte Meyerink, M.S. in Statistics, MSU (RA Spring 2020).

Deborah Pearlman, M.S. in Mathematics Education, MSU (RA in AY 2019–20).

Robin Belton, M.S. in Mathematics, MSU (May 2019).

External PhD Committees

McBeth Ahortor, PhD candidate in Statistics, MSU. Committee member, Nov. 2023–Present.

Yu (Demi) Qin, PhD in CS, Tulane University. Committee member and research mentor/collaborator, 2019–Dec. 2024.

Vladislav Polianskii, KTH Royal Institute of Technology. Grading committee member, Nov. 2022.

Eric Berry, MS in Mathematics, MSU. Committee member, 2016–2018.

Undergraduate RAs

McNair Scholars: Kyla Andersen (2024), Justin O'Dea (2016), Angus Tomlinson (2017).

MSU Emerging Scholars: Maia Grudzien (2016), Jachi Madubuko (2017).

2025 Visiting Summer Student: Lily Rippeteau (REU student from U. Nebraska Lincoln).

2024 Visiting Summer Students: Dwayne McCall II (Leadership Alliance student from Claflin College), Nate Rengo (REU student from Western Colorado University), Vasishta Tumuluri (REU student from UNC Chapel Hill).

2023 Visiting Summer Students: Aakash Dutt (co-advising REU student from U. Maryland), Luke

Padula (REU student from Colorado State), Elijah Smith (Leadership Alliance student from Brooklyn College).

2022 Visiting Summer Students: Dhanush Giriyan (intern from Arizona State U.), Seyong Park (REU student from U. Maryland).

2019 REU Students: Clare DuVal (Clemson U.), Tyler Fallis (Western Washington U.), Emily Flanagan (U. Washington), Luke Askew (Colorado State U.).

2018 REU Students: Thomas Li (UC Berkeley).

2017 REU Students: Maia Grudzien (MSU), Carrington Metts (William & Mary), Kira Wencek (U Rhode Island).

MSU USP Advisees (Undergraduate Research Grants): Kyla Andersen (AY '24–'25), Jo Niglio (AY '24–'25), Maksym Makarchuk (AY '23–'24), Eli Quist (AY '23–'24), Seth Bassetti (AY '20–'21), Mark Braun (Spr. '20), Ben Holmgren* (AYs '18–'20), Brendan Kristiansen* (Spr. '18, Fa. '18), Kevin Browder* (Fa. '18), Ryan Hansen (Sum. '18), Arash Ajam (Spr. '18), Nathan Stouffer (Spr. '18), Jachi Madubuko (AY '17–'18), Rostik Mertz* (AY '16-'17), Hayley Smith (Spr. '17), Angus Tomlinson* (AY '16–'17), Maia Grudzien (Sum. '16), Ryan Thompson (AY '16–'17, Spr. '16), and Sawyer Payne (Spr. '16). (* denotes co-advised with another MSU faculty member).

Other MSU Undergraduate Research Advisees: Lauren Martinez (2021–2022), Kye Frazier-Harris (2021), Daniel Balentin (2020–21), Kendall Black (2018), James Soddy (2017), Saurabh Tulsankar (2018).

Tulane University Undergraduate Research Advisees: Parker Evans (Math, co-advised with C. Wenk, AYs '16–18), Lee Chedister (Math, co-advised with C. Wenk, AY '13–14).

High school mentoring: Abby Hsu, Summer 2022.

Graduate Teaching

Computational Geometry (CSCI 534), Spring 2023. Montana State University, Bozeman, MT.

Computational Topology (CSCI 535), Spring 2018, Spring 2020, Spring 2022, Spring 2024, Fall 2026
Montana State University, Bozeman, MT.

Algorithms (CSCI 532), Spring 2017, Spring 2019
Montana State University, Bozeman, MT.

Computational Geometry and Topology (Special Topics, CSCI 591), Spring 2016
Montana State University, Bozeman, MT.

Topology, September 2011–April 2012
Institute for Science and Technology Austria, Klosterneuburg, Austria.

Undergraduate Teaching

Discrete Structures (CSCI 246), Fall 2018, Spring 2021, Spring 2022 (co-instructor)
Montana State University, Bozeman, MT.

Advanced Algorithmic Topics (CSCI 432), Fall 2015, Fall 2016, Fall 2017, Fall 2019, Fall 2020, Fall 2021, Fall 2022, Fall 2023, Spring 2026
Montana State University, Bozeman, MT.

Special Topics: Computational Topology and Geometry (CSCI 491), Spring 2016
Montana State University, Bozeman, MT.

Computational Geometry, Spring 2013 (co-instructor)
Computer Science Department, Carnegie Mellon University, Pittsburgh, PA.

Discrete Mathematics for Computer Science, Spring 2009 (co-instructor)
Computer Science Department, Duke University, Durham, NC.

Other Teaching

Teaching Assistant, Principles of Computer Science, Spring 2008
Computer Science Department, Duke University, Durham, NC

Student Mentor and Grader, Calculus I and III, January 2004–May 2007
Mathematics and Computer Science Department, Saint Joseph's University, Philadelphia, PA.

Recitation Leader, Business Calculus, Spring 2004
Mathematics and Computer Science Department, Saint Joseph's University, Philadelphia, PA.

Textbook Assistant, *Learning to Program with Alice*, December 2004
Saint Joseph's University, Philadelphia, PA

Awards and Honors

Presidential Early Career Award for Scientists and Engineers (PECASE), January 2025.

10-Year Impact Award, ACM SIGSPATIAL GIS, October 2024. (For paper "Local Persistent Homology Based Distance Between Maps").

Best paper award, IEEE VIS, October 2024: Awarded for the paper "Rapid and Precise Topological Comparison with Merge Tree Neural Networks".

Presenter's Choice Award, 2018: Awarded at the STEM for All Video Showcase for the video "Teaching Computer Science through Storytelling."

Digital Pathology Poster Award, Honorable Mention, 2017: Awarded at SPIE Medical Imaging for the poster "Topological Descriptors for Quantitative Prostate Cancer Morphology Analysis."

Postdoc of the Month, Journal of Postdoctoral Research, January 2016.

Graduate Aid in Areas of National Need Fellowship, August 2007–May 2011.

Barry M. Goldwater Scholarship, 2006.

Google Anita Borg Scholarship Finalist, 2007.

USA Today All-USA College Academic Team, honorable mention, 2005.

Honor Societies: Phi Beta Kappa, Upsilon Pi Epsilon (CS), Pi Mu Epsilon (Math), Sigma Xi

University Awards

Founder's Day Award for Student Excellence (along with student awardee Eli Quist), MSU, 2024.

Excellence in Research Award, College of Engineering, MSU, 2022.

Faculty Excellence Grants (FEG): 2017, 2018, 2024.

Excellence in Outreach Award, College of Engineering, MSU, 2018.

Departmental Research Award, School of Computing, MSU, 2017.

Nominated, CFE Teaching Award, MSU, 2016.

Duke University Computer Science Award for Teaching Discrete Mathematics, Spring 2009

Graduated Summa Cum Laude, B.S., Saint Joseph's University, May 2007.

Saint Joseph's University Mathematics Award, May 2007.

University Scholar, Saint Joseph's University, 2006–2007. Freed from senior year course requirements in order to complete an independent project of unusual breadth, depth and originality. One of two selected to among 764 seniors).

Presidential Scholarship and Del Marco Scholarship, Saint Joseph's University, 2003–2007.

Diocesan Scholar Fellowship, Archdioceses of Philadelphia, 2002–2003.

National Advisee Awards

Anna Schenfisch (Ph.D. Student, Mathematics), 2019 NSF GRFP recipient.

Robin Belton (Ph.D. Student, Mathematics), 2018 NSF GRFP recipient.

Parker Evans (B.S. Student, Mathematics, Tulane U.), 2017 Goldwater scholar.

Campus-Wide Advisee Awards

Eli Quist (B.S. Student, Applied Math), Founder's Day Student Excellence Award, 2024.

Samuel Micka (Ph.D. Student, Computer Science), MSU Dissertation Completion Award, 2020.

Book Chapters

4. L. Fajstrup, BTF, W. Li, L. Mezrag, T. Rask, F. Tombari, and Z. Urbančič. Gromov–Hausdorff Distance for Directed Spaces. AWM Springer Series, special issue on Research in Computational Topology 3, 24 pages, accepted and in press.
3. R. Belton, R. Brooks, S. Ebli, L. Fajstrup, BTF, N. Sanderson, and E. Vidaurre. Combinatorial Conditions for Directed Collapsing. AWM Springer Series, volume 30, special issue on Research in Computational Topology 2, pp. 167–189, Jan. 2022. Also available at arXiv:1912.12759.
2. R. L. Belton, R. Brooks, S. Ebli, L. Fajstrup, BTF, C. Ray, N. Sanderson, and E. Vidaurre. Towards Directed Collapsibility. AWM Springer Series, special issue on AWM Research Symposium, 2020. Also available at arXiv:1902.01039.
1. A. Bittner, BTF, M. Grudzien*, S. Ghosh Hajra, J. Huang, K. Pelatt, C. Thatcher, A. Tumurbaatar, and C. Wenk. Comparing Directed and Weighted Road Maps. AWM-IMA Special Issue on Research in Computational Topology, pp.57–70, Springer, 2018.

Journal Papers

23. B. Christensen, J. Ruder, A. McCleary, BTF, M. Owkes, Geometric and Topological Data Analysis of Atomization Simulations for Model Development. Computers and Fluids, accepted.
22. BTF, S. Micka, D. L. Millman, A. Schenfisch, and L. Williams. A Faithful Discretization of Verbose Directional Transforms. Discrete & Computational Geometry (DCG), 44 pages, April 2026 (published online Nov. 2025). Also available at arXiv:1912.12759.

*Undergraduate student at time of research or submission.

21. Y. Qin, BTF, C. Wenk, and B. Summa. Rapid and Precise Topological Comparison with Merge Tree Neural Networks. IEEE Transactions on Visualization and Computer Graphics (VIS2024). October 2024. Also available at arxiv:2404.05879. 23.2% acceptance rate. Selected as one of five best papers among 124 accepted papers.

20. BTF and A. Patel. Combinatorial Persistent Homology Transform. Foundations of Data Science, Sept. 2024, 6(3): 379-394. DOI: doi: 10.3934/fods.2024020. Also available at arXiv:1707.02251.

19. R. Belton, B. Cummins, BTF, and T. Gedeon. Extremal Event Graphs: A (Stable) Tool for Analyzing Noisy Time Series Data. Foundations of Data Science, Mar. 2023, 5(1): 81-151. DOI: doi: 10.3934/fods.2022019.

18. M. Buchin, E. Chambers, P. Fang*, BTF, E. Gasparovic, E. Munch, and C. Wenk. Distances Between Immersed Graphs: Metric Properties. La Matematica, Vol 2, pp. 197–222, January 2023. DOI: 10.1007/s44007-022-00037-8.

17. BTF, R. Komendarczyk, S. Majhi, and C. Wenk. On the Reconstruction of Geodesic Subspaces of $\mathbb{R}^n$. International Journal of Computational Geometry & Applications, vol. 32, no. 01n02, pp. 91–117. Sept. 2022. Also available at arXiv:1810.10144.

16. J. Cisewski-Kehe, BTF, W. Hellwing, P. Drozda, M.R. Lovell, and M. Wu*. Differentiating Small-Scale Subhalo Distributions in CDM and WDM Models Using Persistent Homology. Physical Review D, vol. 26, no. 2, July 2022.

15. A. Myers, F. Khasawneh, and BTF. ANAPT: Additive Noise Analysis for Persistence Thresholding. Foundations of Data Science 4(2), pp. 243–269. June 2022. http://dx.doi.org/10.3934/fods.2022005. Also available at arXiv:2012.04039.

14. Y. Qin, BTF, C. Wenk, and B. Summa. A Domain-Oblivious Approach for Learning Concise Representations of Filtered Topological Spaces. IEEE Transactions on Visualization and Computer Graphics (VIS2021). October 2021. Also available at arxiv:2105.12208

13. R. L. Belton, BTF, R. Mertz*, S. Micka, D. L. Millman, D. Salinas, A. Schenfisch, J. Schupbach, and L. Williams. Reconstructing Embedded Graphs from Persistence Diagrams. Computational Geometry: Theory and Applications, Vo. 90. Elsevier, Oct. 2020.

12. M. J. Catanzaro, J. M. Curry, BTF, J. Lazovskis, G. Malen, H. Riess, B. Wang, and M. Zabka. Moduli Spaces of Morse Functions for Persistence. Journal of Applied and Computational Topology (APCT), online first. Springer, June 2020.

11. E. Berry, Y.-C. Chen, J. Cisewski-Kehe, and BTF. Functional Summaries of Persistence Diagrams. Journal of Applied and Computational Topology (APCT), Vol. 4, Issue 1. Springer, March 2020. Also available at arXiv:1804.01618.

10. E. Chambers, BTF, Y. Wang. and C. Wenk. Map-Matching Using Shortest Paths. Transactions on Spatial Algorithms and Systems (TSAS), Vol. 6, No. 1, Article no. 6. ACM, Feb. 2020.

9. P. Lawson, A. Sholl, J. Q. Brown, BTF, and C. Wenk. Persistent Homology for the Quantitative Evaluation of Architectural Features in Prostate Cancer Histology. Scientific Reports 9, Article 1139, Feb. 2019.

8. F. Chazal, BTF, F. Lecci, B. Michel, A. Rinaldo, and L. Wasserman. Robust Topological Inference: Distance to a Measure and Kernel Distance. Journal of Machine Learning Research (JMLR), 2018. Also available at arXiv:1412.7197.

7. BTF. Statistical Techniques in TDA with Applications to Real Data. Postdoc Journal 4 (1), Jan. 2016.

6. M. Ahmed, BTF, K. S. Hickmann, and C. Wenk. A Path-Based Distance for Street Map Comparison. ACM TSAS Inaugural Volume 1 (1). Article 3, 28 pages. Aug. 2015.

5. T. Roman, A. Nayyeri, BTF, and R. Schwartz. A Simplicial Complex-Based Approach to Unmixing Tumor Progression Data. BMC Bioinformatics 16, Article no. 254, Aug. 2015.

4. BTF, F. Lecci, A. Rinaldo, L. Wasserman, S. Balakrishnan, and A. Singh. Confidence Sets for Persistence Diagrams. *Annals of Statistics*, vol. 42 (6), pages 2301–2339, 2014. Also available at arXiv:1303.7117.
3. H. Edelsbrunner, BTF, and G. Rote. Add Isotropic Gaussian Kernels at Own Risk: More and More Resilient Modes in Higher Dimensions. *Discrete and Comp. Geom.*, vol. 49 (4), pages 797–822, June 2013.
2. F. Chazal, BTF, F. Lecci, A. Rinaldo, A. Singh, and L. Wasserman. On the Bootstrap for Persistence Diagrams and Landscapes. *Modeling and Analysis of Information Systems*, vol. 20 (6), pages 96–105, Dec. 2013. Also available at arXiv:1311.0376.
1. BTF. The Difference in Lengths of Curves in $\mathbb{R}^n$. *Acta Sci. Math. (Szeged).*, vol. 77 (1-2), pages 359–67, 2011.

Peer-Reviewed Conference Papers

25. J. Cisewski-Kehe, BTF, A. McCleary, E. Quist. Tensor Computation of Euler Characteristic Functions and Transforms. SoCG 2026, to appear. Also available at arXiv:2511.03909.
24. BTF, D. L. Millman, and A. Schenfisch. How Small Can Faithful Sets Be? Ordering Topological Descriptors *Proc. CCCG*, Aug. 2024. 79.5% acceptance rate.
23. J. Chipps, BTF, S. A. Hancock, and B. McCoy. Ant and Bear Dance for Dokweebah: Using a Skokomish Story to Engage Middle School Students in Event-Driven Programming. *Proc. CompEd*, pp. 43–49, Dec. 2023.
22. B. do Amaral, BTF, O. Firth, S. A. Hancock, P. Jeffers, B. Z. Komlos, B. McCoy, and S. Windchief. Integrating Computer Science into Middle School Curricula Through Storytelling: A Lesson Plan on Beaded Bags of the Columbia Plateau. *Proc. CompEd*, pp. 50–56, Dec. 2023.
21. E. Chambers, BTF, B. Holmgren, S. Majhi, and C. Wenk. Metric and Path-Connectedness Properties of the Fréchet Distance for Paths and Graphs. *Proc. CCCG*, pp. 213–224, 2023.
20. H.-C. Chang, BTF, B. McCoy, D. L. Millman, and C. Wenk. From Curves to Words and Back Again: Geometric Computation of Minimum-Area Homotopy. *Proc. WADS*, pp. 605–619, Aug. 2023.
19. K. Buchin, BTF, E. Hosseini Sereshgi, and C. Wenk. On Length-Sensitive Fréchet Similarity. *Proc. WADS*, pp. 208–231, Aug. 2023.
18. BTF, S. Micka, D. L. Millman, A. Schenfisch, and L. Williams. Efficient Graph Reconstruction and Representation Using Augmented Persistence Diagrams. *Proc. CCCG*, pp. 284–292, Aug. 2022.
17. BTF, B. Holmgren*, B. McCoy, and D. L. Millman. If You Must Choose Among Your Children, Pick the Right One. *Proc. CCCG*, Aug. 2020.
16. P. Evans*, BTF, C. Wenk. Combinatorial Properties of Self-Overlapping Curves and Interior Boundaries. *Proc.* 36th *Annu. Symp. Comp. Geo.*, June 2020. 34.1% acceptance rate.
15. BTF, S. Hancock, B. Komlos, B. Kristiansen*, S. Micka and A. Theobold. Bring the Page to Life: Engaging Rural Students in Computer Science Using Alice. *Proc. ITiCSE*, June 2020. 27.6% acceptance rate.
14. P. Lawson, B. Hu, BTF, B. Summa, C. Wenk, and J.Q. Brown. Assessment of Sampling Adequacy Using Persistent Homology for the Evaluation of Heterogeneity in 3D Histology Acquired Through Inverted Selective Plane Illumination Microscopy (iSPIM). Proc. SPIE 11073, Clinical and Preclinical Optical Diagnostics II: 1107316, (SPIE European Conferences on Biomedical Optics (ECBO), invited paper, 5 pages), 2019.
13. R. L. Belton, BTF, R. Mertz*, S. Micka, D. L. Millman, D. Salinas, A. Schenfisch, J. Schupbach, and L. Williams. Learning Simplicial Complexes from Persistence Diagrams. *Proc. CCCG*, Aug. 2018.

12. E. Chambers, BTF, Y. Wang, and C. Wenk. Map-Matching Using Shortest Paths. *Proc. IWISC*, Apr. 2018.
11. K. Buchin, M. Buchin, D. Duran, BTF, R. Jacobs, V. Sacristán, R. I. Silveira, F. Staals and C. Wenk. Clustering Trajectories for Map Construction. *Proc. ACM SIGSPATIAL GIS*, article 14 (10 pages), Nov. 2017. 17.9% acceptance rate.
10. S. Micka, S. Yaw, BTF, B. Mumey and M. P. Wittie. Efficient Multipath Flow Monitoring. *Proc. IFIP Networking*, June 2017. 28.7% acceptance rate.
9. U. Goel, C. Cooper*, BTF, and M. P. Wittie. A First Look at Web Browsing Predictions Using DNS Logs. *Proc.* 25th *Int. Conf. on Software Eng. and Data Eng. (SEDE)*, Sept. 2016, pp. 53–60.
8. BTF, and B. Wang. Exploring Persistent Local Homology in Topological Data Analysis. *Proc. IEEE Int. Conf. Acoustics, Speech, and Signal Processing (ICASSP)*, Mar. 2016, pp. 6430–6434. 49.7% acceptance rate.
7. M. Ahmed, BTF, M. Gibson, and C. Wenk. Choosing Thresholds for Density-Based Map Construction Algorithms. *Proc. ACM SIGSPATIAL GIS*, pp. 43–52, Nov. 2015. 17.9% acceptance rate.
6. M. Cohen*, BTF, G. L. Miller, A. Nayyeri, D. Sheehy, and A. Velingker. Approximating Nearest Neighbor Distances. *Proc. WADS*, Aug. 2015. 36.5% acceptance rate.
5. F. Chazal, BTF, F. Lecci, B. Michel, A. Rinaldo, and L. Wasserman, Subsampling Methods for Persistent Homology. *Proc. ICML*, Jul. 2015. Also available at arXiv:1406.1901. 26% acceptance rate.
4. M. Ahmed, BTF, and C. Wenk. Local Persistent Homology Based Distance Between Maps. *Proc. ACM SIGSPATIAL GIS*, Nov. 2014. 21.2% acceptance rate. (10-year impact award in 2024).
3. F. Chazal, BTF, F. Lecci, A. Rinaldo, and L. Wasserman. Stochastic Convergence of Persistence Landscapes and Silhouettes. *Proc.* 27th *Annu. Symp. Comp. Geo.*, pages 474–83, June 2014. Also available at arXiv:1313.0308. 34% acceptance rate.
2. M. Cohen*, BTF, G. L. Miller, A. Nayyeri, R. Peng, and N. Walkington. Solving 1-Laplacians in Nearly Linear Time: Collapsing and Expanding a Topological Ball. *Proc. ACM-SIAM Symposium on Discrete Algorithms*, pages 204–16, January 2014. 28% acceptance rate.
1. H. Edelsbrunner, BTF, and G. Rote. Add Isotropic Gaussian Kernels at Own Risk: More and More Resilient Modes in Higher Dimensions. *Proc. of the* 27th *Annu. Symp. Comp. Geo.*, pages 91–100, June 2012. 35% acceptance rate.

Invited Articles

2. BTF and B. Wang. Open Problems Column, ACM SIGACT News 48 (1), Aug. 2017.
1. E. Chambers, BTF, and L. Ziegelmeier. WinCompTop 2016. Association for Women in Mathematics (AWM) Newsletter 46(6), Nov. 2016.

Workshop Contributions, Extended Abstracts, and Short Papers

19. J. Sriraman, E. Barton, BTF, D. L. Millman, B. Mumey, N. Rengo*, B. Sopp, V. Tumuluri*, and B. Zhu. Parametric Shortest Paths in a Linearly Interpolated Graph. *Computational Geometry: Young Researcher's Forum (YRF)*, 2026.
18. B. V. Christensen, M. Owkes, J. Ruder, A. McCleary, and BTF. Efficient Geometric and Topological Data Analysis of Atomization Simulations. *Conference on Liquid Atomization and Spray Systems*. Paper 49, four pages. May 2025.
17. E. Chambers, BTF, E. Hosseini Sereshgi, M. Löffler. Drawing Reeb Graphs. *36th International Workshop on Combinatorial Algorithms (IWOCA 2025)*, pages 58–71, July 2025. 45% acceptance rate.
16. D. Munoz, E. Munch, BTF, F. A. Khasawneh. The Walk-Length Filtration for Persistent Homology on Weighted Directed Graphs. *Computational Geometry: Young Researcher's Forum (YRF)*, 2024.

15. BTF, D. L. Millman, and A. Schenfisch. Lower Bounding Minimal Faithful Sets of Verbose Persistence Diagrams. *The European Workshop on Computational Geometry (EuroCG)*, March 2024.
14. Y. Qin, BTF, C. Wenk, and B. Summa. Visualizing Topological Importance: A Class-Driven Approach. *TopoInViz, a Workshop in Conjunction with IEEE VIS*, 2023.
13. H.-C. Chang, BTF, B. McCoy, D. L. Millman, and C. Wenk. From Curves to Words. *Computational Geometry: Young Researcher's Forum (YRF)*, pp. 27–30, 2023.
12. D. Giriyan*, J. Cisewski-Kehe, and BTF. Image Shape Classification with the Weighted Euler Curve Transform. *Computational Geometry: Young Researcher's Forum (YRF)*, pp. 41–44, 2023.
11. W. DuBow, D. Arnold, BTF, and M. Gutierrez Soto. Challenges and Successes in Writing BPC Plans for NSF Proposals. *SiGCSE Panel*, 2023.
10. E. Chambers, BTF, B. Holmgren*, S. Majhi, and C. Wenk. Path-Connectivity of Fréchet Spaces of Graphs. *Computational Geometry: Young Researchers Forum*, 2022.
9. BTF, D. L. Millman, E. Pryor*, and N. Stouffer*. DBSpan: Density-Based Spanner for Clustering Complex Data, With an Application to Persistence Diagrams. Applications of Topological Data Analysis to Data Science, Artificial Intelligence, and Machine Learning (TDA at SDM). April 2022. 6 pages.
8. H. Bratterud, M. Burgess, BTF, D. L. Millman, T. Oster*, and C. Sung. The Sung Diagram: Revitalizing the Eisenhower Matrix (poster paper). Diagrams 2020.
7. BTF, S. Micka, D. L. Millman, A. Schenfisch, and L. Williams. Challenges in Reconstructing Shapes from Euler Characteristic Curves. *Fall Workshop on Computational Geometry*, 2018.
6. BTF, S. Majhi and C. Wenk. Threshold-Based Graph Reconstruction Using Discrete Morse Theory. *Fall Workshop on Computational Geometry*, 2018.
5. A. Abdelkader, G. Boeing, BTF, and D. L. Millman Topological Distance Between Nonplanar Transportation Networks. *Fall Workshop on Computational Geometry*, 2018.
4. A. Schenfisch, and BTF. Curvature Estimates of Point Clouds as a Tool in Quantitative Prostate Cancer Classification. *Computational Geometry: Young Researchers Forum*, 2018.
3. P.J. Lawson, B. Hu, BTF, C. Wenk, J.Q. Brown. Quantifying Prostate Cancer Morphology in 3D Using Light Sheet Microscopy and Persistent Homology (Conference Presentation). Proc. SPIE 10472, Diagnosis and Treatment of Diseases in the Breast and Reproductive System IV: 1047209, (SPIE BiOS), Jan. 2018. doi: 10.1117/12.2290994. 22% acceptance rate.
2. T. Roman, BTF, A. Nayyeri, G. L. Miller and R. Schwartz. Improved Geometric Unmixing Models for Tumor Progression. *Proc. Great Lakes Bioinformatics Conf.*, May 2014.
1. T. Roman, BTF, A. Nayyeri, G. L. Miller and R. Schwartz. Determining Low-Dimensional Embeddings in High-Dimensional Genotype Space for Tumor Phylogeny Reconstruction. *Proc. Great Lakes Bioinformatics Conf.*, May 2013.

Works in Progress

9. BTF, M. Makarchuk*, S. Micka, and D. L. Millman. Too Many or Too Few? SBounds for Topological Descriptor In submission.
8. BTF, X. He, Z. Liu, , D. L. Millman, and B. Zhu. Approximate Nearest Nein the Space of Persistence Diagrams. In submission. Available at arXiv:1812.11257.
7. J. Cisewski-Kehe, BTF, D. Giriyan*, and E. Quist*. The Expected Weighted Euler Characteristic Transform for Images. In preparation.
6. BTF, S. Fox, S. Schenfisch, J. Schupbach, and E. Quist*. Simulation of Prostate Cancer Histology Slides and Glands. In preparation.
5. BTF, B. Grossenbacher-Rieck, B. Holmgren, E. Quist, and J. Schupbach. Intrinsic Validation of Manifold Learning Algorithms. In preparation.

4. BTF, B. Holmgren, B. McCoy, D. L. Millman, J. Niglio*, and B. Zhu. On Computing Discrete Morse Functions. In preparation.
3. BTF, S. Nandi, and B. Sopp. Persistence Diagrams as a Spatial Descriptor. In preparation.
2. E. Wolf Chambers, BTF, B. Holmgren, and Y. Qin. A Geometric Distance for Morse–Smale Complexes. In preparation.
1. BTF, S. Karakoç, C. Wenk. On Minimum Area Homotopies of Normal Curves in the Plane. In preparation. Available at arXiv:1707.02251.

Other Manuscripts

5. BTF, J. Kim, F. Lecci, C. Maria, and V. Rouveau. Introduction to the R Package TDA. 2014. Available at arXiv:1411.1830.
4. BTF. Modes of Gaussian Mixtures and an Inequality for the Distance Between Curves in Space. PhD Dissertation, Duke University, Durham, NC. June 2012.
3. BTF. The Total Curvature of a Knotted Curve. Translation of Istvan Fáry's Sur la courbure totale d'une courbe gauche faisant un noeud. 2010. http://www.cs.duke.edu/~brittany/research/fary.pdf.
2. BTF. Persistence Diagrams and the Heat Equation Homotopy. 2010. Available at arXiv:1002.1937.
1. BTF. Homotopy Classification of the Components of the Space of Maps into an Aspherical Space: a Problem in the Intersection of Group Theory and Topology. Undergraduate Honors Thesis, Saint Joseph's University, Philadelphia, PA. May 2003.

Software

pyECT. This package provides an efficient and simple implementation of the WECT using PyTorch. Code accompanies SoCG 2026 paper "Tensor Computation of Euler Characteristic Functions and Transforms." https://github.com/compTAG/pyECT.

TDA: Statistical Tools for Topological Data Analysis, R Package. Co-developed with Fabrizio Lecci in 2014. Maintained by BTF, J. Kim, C. Maria, D. L. Millman, and V. Rouvreau http://cran.r-project.org/web/packages/TDA/.

Videos

Teaching Computer Science through Storytelling, 2018 STEM for All Video Showcase. http://stemforall2018.videohall.com/presentations/1293. **Presenter's Choice Award.**

Book Reviews

Statistical Analysis of Contingency Tables by M.W. Fagerland, S. Lydersen, P. Laake. Review by A. Schenfisch and BTF. *The American Statistician* 73(2), p. 208, 2019.

The Structure and Stability of Persistence Modules by F. Chazal, V. de Silva, M. Glisse, and S. Oudot. Review by R. Belton and BTF. ACM SIGACT News 48 (2), pp.7–11, June 2017.

Polyhedral and Algebraic Methods in Computational Geometry by M. Joswig and T. Theobald. Review by BTF and D. L. Millman. ACM SIGACT News, 2015.

Discrete and Computational Geometry by S.L. Devadoss and J. O'Rourke. ACM SIGACT News 45(1), Mar. 2014.

How to Fold It by J. O'Rourke. Review by BTF and David L. Millman. ACM SIGACT News 44(3), Sept. 2013.

Geometric Algebra: an Algebraic System for Computer Games and Animation by J. Vince. Review by BTF and D. L. Millman, ACM SIGACT News, March 2011.

Geometric Folding Algorithms: Linkages, Origami, Polyhedra by E. D. Demaine and J. O'Rourke. Review by BTF and D. L. Millman. ACM SIGACT News 42(1), March 2011.

Higher Arithmetic: An Algorithmic Introduction to Number Theory by H. Edwards. Review by BTF and D. L. Millman. ACM SIGACT News 40(2), June 2009.

Geometric Algebra for Computer Science by L. Dorst, D. Fontijne, and S. Mann. Review by BTF and D. L. Millman. ACM SIGACT News 39(4), Dec. 2008.

Conference Paper Presentations

Stochastic Convergence of Persistence Landscapes and Silhouettes. Annu. Symp. Comp. Geo. Kyoto U., Japan. June 2014.

Add Isotropic Gaussian Kernels at Own Risk. Annu. Symp. Comp. Geo. U. North Carolina, Chapel Hill, NC, June 2012. **Runner up for best student presentation award.**

Conference & Workshop Invited Presentations

Topological Descriptors: Theory and Practice. Special Session on Advances in Applied Topology: Theory and Applications, Spring Eastern AMS Sectional Meeting. October 2026.

How Strong Are Topological Descriptors? Special session on Applied and Computational Topology, Joint Mathematics Meetings of AustMS, NZMS, and AMS, Auckland, New Zealand, December 2024.

Topological Descriptors for Large Data. Workshop on Applications of Topological Data Analysis to Mathematical Biology, MATRIX Institute, Creswick, Australia, November 2024.

Faithful Discretizations of the PHT Workshop on Computational Topology and Geometry, Foundations of Computational Mathematics (FOCM), Paris, June 2023.

Comparing and Analyzing Sets of Persistence Diagrams. Workshop on Computational Topology and Applications, Spring Western AMS Sectional Meeting. May 2022.

Working with Persistence Diagrams: Some Theory and Some Practical Tips/Tricks. Bridging Applied and Quantitative Topology, online workshop. May 2022.

Distances and Near Neighbors in Spaces of Persistence Diagrams. Special Session on "Statistics and Machine Learning using Topology and Geometry", Joint Mathematics Meeting. April 2022. (Postponed from Jan. 2022).

Topological Descriptors: A Cellular View. Special Session on Algebraic Combinatorics and Category Theory in Topological Data Analysis, AMS Spring Southeastern Sectional Meeting. March 2022. Canceled.

Searching in the Space of Persistence Diagrams. Topological Data Analysis and Beyond, Neurips Workshop (virtual). Dec. 2020.

Panelist, *What Do We Mean When We Say We're Being Culturally Responsive?* Association for Educational Communications and Technology (AECT) Conference (virtual). Nov. 2020.

Fostering Collaboration among Research Groups. AISES Lighting the Pathway to Faculty Careers for Natives in STEM" (LTP). Webinar. Aug. 2020.

Finite Representations of Shapes in Euclidean Space. Optimal Transport, Topological Data Analysis and Applications to Shape and Machine Learning. July 2020.

Inverse Problems with Persistence. Special Session on Applied Topology JMM. Jan. 2020.

Topology Tutorial, an invited two-part lecture series at ICERM workshop, *Applied Mathematical*

Modeling with Topological Techniques, Aug. 2019.

LPH on Real Roads. AWM Workshop for Women in Applied and Computational Topology, JMM. Jan. 2019.

Topological Data Analysis: Roads and Histology. Machine Learning Workshop, CG Week. Budapest, Hungary. June 2018.

Locality-Sensitive Searching in the Space of Persistence Diagrams. Abel Symposium. Geiranger, Norway. June 2018.

Topological Descriptors for Data Comparison SIAM SEAS, Chapel Hill, NC. March 2018.

Understanding and Comparing Data Sets Using Topological Descriptors. Hausdorff Institute for Mathematics, Bonn, Germany. May 2017.

Storytelling to Improve the Pipeline to CS. Big Ideas in Big Sky; Big Sky, MT. May 2017.

Topological Descriptors. AMS Special Session on Statistical Methods in Computational Topology and Applications, Joint Mathematics Meeting. Atlanta, GA. January 2017.

Using Topological Data Analysis to Study Glandular Architecture. 1st International Workshop on Topological Data Analysis in Biomedicine (TDA-Bio), ACM BCB, Seattle, WA. October 2016.

Road Network Comparison: an Application of Topological Data Analysis. XXIst Oporto meeting on Geometry, Topology, and Physics; session on Applications of Topology. Lisbon, Portugal. February 2015.

Map Construction and Comparison Using Local Structure. Workshop on Topological Data Analysis. SAMSI, Research Triangle Park, NC. February 2014.

The Intersection of Statistics and Topology. AWM Workshop for Women Graduate Students and Recent PhDs, Joint Mathematics Meetings, January 2014.

Modes of Gaussian Mixtures. Workshop on Topological Data Analysis and Machine Learning Theory, Banff International Research Station, October 2012.

Geometry of Gaussian Mixtures. Yaroslavl' International Conf. Discrete Geometry. Yaroslavl', Russia. August 2012.

Persistence of Extra Modes in Gaussian Mixtures. Workshop on Computational Topology at CG Week 2012, U. North Carolina, Chapel Hill, NC. June 2012.

Finding Ghost Features in Gaussian Mixture Models. Workshop on Sphere Arrangements. Field's Institute, Toronto, Canada, Nov. 2011.

Computing with Alice. SIGCSE Kid's Camp, Milwaukee, WI, March 2010.

Contributed Presentations

Panelist, *Challenges and Successes in Writing BPC Plans for NSF Proposals.* Co-panelists D. Arnold and M. Gutierez Soto. Moderator W. DuBow. SiGCSE, March 2023.

Directed Homotopy Collapses, Women in Topology research session at AWM Research Symposium, April 2019.

Persistence of Wild Type. HerbertFest. Klosterneuberg, Austria. June 2018.

Teaching Computational Geometry and Topology. Educational Forum, CG Week. Budapest, Hungary. June 2018.

Topological Descriptors: Statistics and Applications. Applied Algebraic Topology. Sapporo, Japan. August 2017.

Using Statistics in Topological Data Analysis. AWM Research Symposium. College Park, MD. April 2015.

Statistical Inference for Persistent Homology. International Conf. Geometry, Topology, and Applications, Yaroslavl', Russia. Sept. 2013.

Statistical Inference for Persistent Homology. Fifth Discrete Geometry and Algebraic Combinatorics Conf., Brownsville, TX, April 2013.

A Geometric View of Gaussian Mixtures. Young Researcher's Forum at CG Week 2012, U. North Carolina, Chapel Hill, NC. June 2012.

Understanding Isotropic Gaussian Mixture Models. Discrete and Computational Geometry and Topology Workshop. IST Austria, Klosterneuburg, Austria. March 2012.

Exploring Computational Mathematics: Unfolding Polyhedra. Joint presentation with D. L. Millman. Contributed Paper Session, MathFest. Madison, WI, August 2008.

Realizability Problems in Group Theory. Spring Meeting of the EPADEL Section of the Mathematics Association of America. Shippensburg State U., April 2006.

Colloquia, Seminars, and Guest Lectures

TDA With Real Data: Un-Complicating Distance Computations. Computational Topology Class guest lecture, U. Utah, April 2026.

Representing Shapes with Topological Descriptors. Algorithms Seminar, Duke U., 16 October 2024.

Topological Descriptors for Data Comparison. Departmental Seminar, Villanova U., October 2024.

Describing Shapes with Topological Descriptors. TDA Seminar, Michigan State U., September 2024.

Topological Descriptors and Invariants: Describing Differences. Macalester College Mathematics Seminar, Macalester College, September 2024.

Topology and Describing Differences. Computational Geometry Class guest lecture, Western Colorado U., April 2024.

TDA in Graph Comparison. TDA Class guest lecture, U. Utah, April 2021.

Finite Representations of Shapes in Euclidean Space. Michigan State U., November 2020.

Topological Descriptors, Mathematics Seminar, U. Delaware, May 2018.

Topological Descriptors for Data Comparison, Engineering Seminar, Arizona State U., March 2018.

Applied Algebraic Topology: Integrating Math, Statistics, Computer Science, and Applications. U. Puget Sound, Tacoma, WA. Oct. 2016.

Using TDA in Prostate Cancer Diagnosis and Prognosis. Applied Math Seminar, Colorado State U. Fort Collins, CO. Oct. 2016.

Persistent Local Homology in Road Network Analysis. Topology Seminar, Pomona College. Claremont, CA. Oct. 2016.

Finding Shape in Data. Claremont Colleges Mathematics Colloquia (CCMC). Claremont, CA. Oct. 2016.

The Intersection of Statistics and Topology. Institute of Statistical Mathematics. Tachikawa, Japan. June 2014.

Local Homology Based Distance Between Embedded Graphs. Active Robotic Sensing Lab. North Carolina State U., Raleigh, NC. March 2014.

Local Homology Based Distance Between Embedded Graphs. Data Seminar. Duke U., Durham, NC. March 2014.

From Point Clouds to Computations: How to Understand Data from the Perspective of a Topologist. Rhodes U., Memphis, TN. October 2013.

Gaussian Mixtures: A Common Model with Unexpected Features. Department of Mathematics, Bremen U., Bremen, Germany. May 2013.

Counting and Locating Ghost Modes in a Gaussian Mixture. Department of Mathematics, U. Pennsylvania, Philadelphia, PA. January 2013.

Metrics on Persistence Diagrams. Discrete and Computational Geometry Summer School. Yaroslavl', Russia. July 2012.

Ghosts in Gaussian Mixture Models. Data Seminar. Duke U., Durham, NC. January 2012.

Finding Ghost Features in Gaussian Mixture Models. CMU Theory Lunch. Carnegie Mellon U., Pittsburgh, PA. Nov. 2011.

Finding Ghost Features in Gaussian Mixture Models. Mathematics and Computer Science Colloquium. Saint Joseph's U., Philadelphia, PA. Nov. 2011.

What is Computational Topology? Mathematics and Computer Science Colloquium. Saint Joseph's U., Philadelphia, PA. December, 2008.

Realizability Problems in Group Theory. Joint presentation with Julia Fox. Research Seminar, Program for Women in Mathematics. Institute for Advanced Study, Princeton, NJ, May 2006.

Campus Talks

Introduction to TDA: A Biased Crash-Course. Bozeman Ecology and Environmental Statistics Group Meeting, April 2022.

Bringing Research to the Community ... and Across Campus. CBME Webinar Series, Jan. 2022.

Directional Topological Descriptors for Shape Representation. Mathematics Seminar, MSU, Bozeman, MT. March 2021.

Computational Topology and CS Outreach. Honors Seminar, MSU, Bozeman, MT. Oct. 2018.

Studying Data Through the Lens of (Persistent) Homology. Applied Math Seminar, MSU, Bozeman, MT. Nov. 2016.

A Biased Introduction to Computational Topology. Graduate Student Colloquium. Mathematics Department, Tulane U., New Orleans, LA. March 2014.

Persistent Homology. Topology Seminar. Mathematics Department, Tulane U., New Orleans, LA. Sept. 2013.

Modes of Gaussian Mixtures. CMU Statistical Machine Learning Reading Group, Carnegie Mellon U., Pittsburgh, PA. October 2012.

Realizability Problems in Group Theory. Mathematics Awareness Day. Saint Joseph's U., Philadelphia, PA. March 2006.

Teaching with Alice. Joint presentation with John Paul Craig. Summer Research Seminar. Saint Joseph's U., Philadelphia, PA. August 2004 and 2005.

Alice. Joint presentation with John Paul Craig. Mathematics Awareness Day, Saint Joseph's U.. March 2005.

National and International Poster Presentations

Lily Rippeteau (presenter) and BTF. *The Road to Efficient Graph Comparison: Assessing the DIWECT's Graph Retrieval Success and Speed.* Undergraduate Mathematics Symposium, U. Illinois at Chicago, 8 November 2025.

Using Weighted Euler Characteristic Transform for Atomization Model Development. Brendan Christensen, BTF, Alexander McCleary, Mark Owkes, and Jack Ruder (presenter). ATMCS contributed poster session (peer-reviewed extended abstract), ATMCS 2025, pages 154–158.

A Geometric Distance for Morse–Smale Complexes. Benjamin Holmgren (presenter), Erin Wolf Chambers, BTF, and Yu Qin. ATMCS contributed poster session (peer-reviewed extended abstract), ATMCS 2025, pages 204–208.

Rapid and Precise Comparison of Labeled Trees with Neural Networks. Yu Qin, Brittany Terese Fasy, Carola Wenk (presenter), and Brian Summa. ATMCS contributed poster session (peer-reviewed extended abstract), ATMCS 2025, pages 230-245.

Persistence Diagrams as a Spatial Descriptor. Braeden Sopp (presenter), BTF, and Shinjini Nandi. ATMCS contributed poster session (peer-reviewed extended abstract), ATMCS 2025, pages 264-270.

Persistence Diagram Valued Random Fields. Co-authored poster with Elijah Smith, Bradley McCoy, and Braeden Sopp (Presented by Elijah Smith). Leadership Alliance National Symposium, July 2023.

A Domain-Oblivious Approach for Learning Concise Representations of Filtered Topological Spaces for Clustering. Co-authored poster with Yu Qin, Brian Summa, and Carola Wenk. (Presented by Yu Qin). Women in Machine Learning Workshop, NeurIPS 2022.

Culturally Responsive Storytelling Across Content Areas Using American Indian Ledger Art and Physical Computing. Co-authored poster with Jake Chipps (presenter), Stacey Hancock, Aayushi Dangol, Kristin Searle, Colby Tofel-Grehl, Aubrey Rogowski, and Mengying Jiang. ASEE, grantee session poster paper, June 2022.

Comparing Distance Metrics on Vectorized Persistence Summaries. Co-authored poster with Yu

Qin, Brian Summa, and Carola Wenk. (Presented by Yu Qin). Topological Data Analysis Workshop, IMSI (virtual), April 2021.

Using Hasse Diagrams to Compute a Gradient Vector Field. Co-authored poster with Ben Holmgren and David L. Millman. (Presented by Ben Holmgren). NCUR (virtual) , 13 April 2021.

Comparing Distance Metrics on Vectorized Persistence Summaries. Co-authored poster with Yu Qin, Brian Summa, and Carola Wenk. (Presented by Yu Qin). Topological Data Analysis and Beyond, Neurips Workshop (virtual), December 2020.

Directed Collapsibility. Co-authored poster with Robin Belton, Robyn Brooks, Stefania Ebli, Lisbeth Fajstrup, Nicole Sanderson, and Elizabeth Vidaurre. (Presented by Robin Belton). ATMCS, 2020 (postponed due to COVID-19).

Analyzing Musical Compositions with Topological Data Analysis. Co-authored poster with Robin Belton, David L. Millman, Angus Tomlinson*, and Kira Wencek*. (Presented by Robin Belton). ATMCS, Jun. 2018.

Representing Musical Structure with Simplicial Complexes. Co-authored poster with Angus Tomlinson*, Robin Belton, Kira Wencek*. and David L. Millman. (Presented by Angus Tomlinson). ERN, Feb. 2018.

Quantifying Music Complexity Using Topological Data Analysis. Co-authored poster with Kira Wencek*. Robin Belton, Angus Tomlinson*, and David L. Millman. (Presented by Kira Wencek). ERN, Feb. 2018.

American Indian Storytelling with Alice. Co-authored poster with Stacey Hancock, Jachi Madubuko*, Sam Micka, Allison Theobold (Presented by Sam Micka). SIGCSE, Feb. 2018.

Topological Descriptors for Quantitative Prostate Cancer Morphology Analysis. Co-authored poster with Pete Lawson, Eric Berry, J. Quincy Brown, BTF, and Carola Wenk. (Presented by Pete Lawson). Conference on Digital Pathology, SPIE Medical Imaging. Feb. 2017. **Honorable Mention, Digital Pathology Poster Award**

Persistent Homology for Pan-Genome Analysis. Co-authored poster with Alan Cleary, BTF, Thiruvarangan Ramaraj, Joann Mudge, and Brendan Mumey. (Presented by BTF). WinCompTop, Institute for Mathematics and its Applications, Minneapolis, MN. August 2016.

Road Network Analysis. M. Grudzien* and BTF. WinCompTop, Institute for Mathematics and its Applications, Minneapolis, MN. August 2016.

Persistent Homology for Pan-Genome Analysis. Co-authored poster with Alan Cleary, BTF, Thiruvarangan Ramaraj, Joann Mudge, and Brendan Mumey. (Presented by Alan Cleary). Sequencing, Finishing, and Analysis in the Future (SFAF), Santa Fe, NM. June 2016.

On Minimum Area Homotopies. BTF, *S. Karakoç*, C. Wenk, Topology, Geometry, and Data Analysis Conference (TGDA), Ohio State U., May 2016.

Towards an Automated Quantitative Diagnosis of Prostate Cancer. BTF, Q. Brown, P. Lawson, C. Miller, and C. Wenk. (Presented by BTF). BD2K All-Hands Grantee Meeting, NIH, November 2015.

Statistical Inference For Persistent Homology. Joint presentation with F. Lecci. Workshop on Topological Data Analysis, Institute for Mathematics and its Applications, U. Minnesota, Minneapolis,

MN. October 2013.

Homotopy Classification. MathFest, Madison, WI, August 2008.

QShuffler. Joint presentation with Herodotos Herodotou. Graduate Student Research Day Presentation, Duke U., Durham, NC, April 2008.

Visualizing Lists and Arrays. Consortium for Computing Sciences in Colleges Southeastern Region, Lenoir Rhyne College, Hickory, NC, Nov. 2005.

Expanding Alice. Regional Sigma Xi Symposium, Saint Joseph's U., Philadelphia, PA, April 2005.

Campus Poster Presentations

Kyla Anderson (presenter), BTF, Jack Ruder, and Jacob Sriraman. *Exploring How Road Topology Influences Crash Severity via the Weighted Euler Characteristic Transform.* McNair Research Celebration, MSU, December 2025

Kyla Anderson (presenter), Jack Ruder, and BTF. *Investigating the Relationship Between Road Topology and Crash Severity Using the Weighted Euler Characteristic Transform.* Summer Research Celebration, MSU, August 2025.

Lily Rippeteau (presenter) and BTF. *The Road to Efficient Graph Comparison: Assessing the DIWECT's Graph Retrieval Success and Speed.* Summer Research Celebration, MSU, August 2025.

Kyla Andersen (presenter), BTF, Chris Organ, Jack Ruder. *Digital Analysis of Dinosaur Bone Morphology Through 3D Feature Extraction.* McNair Research Celebration, MSU, Dec. 2024.

Benjamin Holmgren, Eli Quist (presenter), Jordan Schupbach, Brittany Terese Fasy, and Bastian Reich. *An Intrinsic Method for the Validation of Machine Learning.* Spring Research Celebration, MSU, April 2024.

Maksym Makarchuk (presenter), Brittany Terese Fasy, Samuel Micka, David L. Millman, and Luke Padula. *Sampling Bounds for Topological Descriptors.* Spring Research Celebration, MSU, April 2024.

Seth Bassetti (presenter), BTF, and David Millman. *Enhancing Accessibility in Topological Data Analysis*, Spring Research Celebration, MSU, April 2021.

Ryan Hansen (presenter) and BTF. *Assessing Model Fitness Using Sheaves.* Student Research Celebration, MSU, April 2019.

Ben Holmgren (presenter), Brittany Terese Fasy, and David L. Millman. *Updating the R Package 'TDA'.* Student Research Celebration, MSU, April 2019.

Jachike Madubuko (presenter) and BTF. *Using Storytelling to Improve the Pipeline for Rural and American Indian Students Entering Computer Science.* Student Research Celebration, MSU, April 2018.

Justin O'Dea (presenter), BTF, and Dave Millman. *Covariance in the Human Brain.* Student Research Celebration, MSU, April 2018.

Nathan Stouffer (presenter) and BTF. *Generating Data for Prostate Cancer Diagnosis.* Student Research Celebration, MSU, April 2018.

Angus Tomlinson (presenter), BTF, and David Millman. *Representing Music with Simplicial Com-*

plexes. Student Research Celebration, MSU, April 2018.

Maia Grudzien (presenter) and BTF. *Road Network Analysis*. Student Research Celebration, MSU, April 2017.

Rostik Mertz (presenter), Brendan Mumey, and BTF. *Bike Networking*. Student Research Celebration, MSU, April 2017.

Justin O'Dea (presenter), BTF, and Dave Millman. *Covariance in the Human Brain*. Student Research Celebration, MSU, April 2017.

Hayley Smith, BTF, Roger Fischer, and Stacey Hancock. *Storytelling: Computer Science for American Indians in the Middle Grades*. Student Research Celebration, MSU, April 2017.

Ryan Thompson (presenter) and BTF. *Applying Techniques from Topological Data Analysis on a Celestial Data Set: The Bigger Picture*. Student Research Celebration, MSU, April 2017.

Angus Tomlinson (presenter), BTF, and David Millman. *An Investigation of Musical Structure with Persistent Homology*. Student Research Celebration, MSU, April 2017.

Sawyer Payne (presenter) and BTF. *Analysis of Prostate Cancer with Persistent Topology and Homology*. Student Research Celebration, MSU, April 2016.

Ryan Thompson (presenter) and BTF. *Applying Techniques from Topological Data Analysis on a Celestial Data Set*. Student Research Celebration, MSU, April 2016.

Student Research Celebration, MSU, April 2016.

Statistical Inference for Persistent Homology. SSE Research Day, School of Science and Engineering, Tulane U., New Orleans, LA. April 2014.

Two Classification Problems in Group Theory and Homotopy Theory. BTF and S. Smith. (Presented by BTF). 18th Annual Student Research Symposium. Saint Joseph's U, Philadelphia, PA, April 2007.

Expanding Alice. Celebration of Student Scholar Activities, Saint Joseph's U., Philadelphia, PA, Sept. 2005.

Alice. Celebration of Student Scholar Activities, Saint Joseph's U., Philadelphia, PA, Sept. 2004.

Statistical Inference for Persistent Homology. SSE Research Day, School of Science and Engineering, Tulane U., New Orleans, LA. April 2014.

Two Classification Problems in Group Theory and Homotopy Theory. BTF and S. Smith. (Presented by BTF). 18th Annual Student Research Symposium. Saint Joseph's U, Philadelphia, PA, April 2007.

Expanding Alice. Celebration of Student Scholar Activities, Saint Joseph's U., Philadelphia, PA, Sept. 2005.

Alice. Celebration of Student Scholar Activities, Saint Joseph's U., Philadelphia, PA, Sept. 2004.

Editorial Boards Guest Editor, Special Collection for Algebraic Topology: Methods, Computation, and Science (ATMCS11), *La Matematica*, Springer, July 2025–Present.

Guest Editor, Special Collection on Advances in Applied and Computational Topology, *La Matematica*, Springer, Dec. 2022–Jan. 2025.

Guest Editor, Special Issue on Algorithmic Aspects of Computational and Applied Topology, *Computational Geometry: Theory & Applications*, Elsevier, 2021–2025.

Editorial Board Member, *Foundations of Data Science*, AIMS, Jan. 2019–Present.

Event Organization

Algebraic Topology: Methods, Computation, and Science (ATMCS11). Co-organizers: BTF and Facundo Memoli. Bozeman, MT, Jul. 2025.

Topology for Data Science (T4DS) Train-the-Trainer. Co-Organizers and Facilitators: BTF and Ben Holmgren. online, June 2023. https://comptag.github.io/t4ds-training

Topology for Data Science (T4DS). Co-organizers: BTF and Ben Holmgren. Bozeman, MT, Apr. 2023. https://comptag.github.io/t4ds

AWM Special Session on Women in Computational Topology, Joint Mathematics meeting (JMM). Co-organizers: BTF and Lori Ziegelmeier. Seattle, Jan. 2022.

Fall Workshop on Computational Geometry. Co-organizers: BTF, D. Millman, and B. Zhu. Virtual, Oct. 2021.

Topological Data Analysis Workshop, Institute for Mathematical and Statistical Innovation (IMSI). Co-organizers: BTF, Kathryn Hess, Matthew Kahle, Sayan Mukherjee, and Jose Perea. April 2021.

Computational Topology research session at AWM Research Symposium, April 2019.

HerbertFest, a celebration of Herbert Edelsbrunner's 60th Birthday. Co-organizers BTF, Dmitriy Morozov, Amit Patel, and Yusu Wang. June 2017.

Applications of Algebraic Topology, minisymposium at the SIAM Central States Meeting. Co-organizers BTF and Lori Ziegelmeier. Colorado State University, Oct. 2017.

Statistics and Applied Algebraic Topology, minisymposium at the SIAM Conference on Applied Algebraic Geometry. Co-organizers BTF and Sayan Mukherjee. Georgia Tech, Aug. 2017.

From Observations to Prediction of Movement. Dagstuhl seminar 17282. Co-organizers Mark Birkin, Somayeh Dodge, BTF, Richard Mann. Germany, Jul. 2017.

Workshop for Women in Computational Topology (WinCompTop). Co-organizers Erin Chambers, BTF, Lori Ziegelmeier. IMA, Minneapolis, MN. Aug. 2016.

Mini-symposium on Computational Topology, workshop during CG Week. Co-organizers BTF, Elizabeth Munch, Don Sheehy. TU Eindhoven, Netherlands, Jun. 2015.

Other Professional Service

MSU Representative, Rocky Mountain Mathematics Consortium. 2019–Present. Served as Secretary in 2022, Vice Chair in 2023, and Chair in 2024.

Program Committee, Fall Workshop on Computational Geometry (FWCG). 2025, 2026.

Program Committee, CG Week: Media Exposition. 2022, 2025.

Advisory Board, Native STEM Portraits. 06/2020–05/2025.

Steering Committee, Women in Computational Topology (WinCompTop). 2016–2024.

Advisory Board, Grace Hopper Conference. 2022–2023.

Scientific Committee, ATMCS. 2020, 2022.

Program Committee. CCCG 2021.

Evaluator, ACM Student Research Competition (SRC) Grand Finals. 2021–22.

Program Committee. SoCG 2021.

Program Committee, CG Week: Young Researcher's Forum (YRF). 2021.

Co-Chair, Poster Committee, Grace Hopper Conference. 2020–2021.

Committee member, Justice, Equity, Diversity, and Inclusion (JEDI) Outreach group, American Statistical Association. Summer 2020–2024.

Diversity Initiative Task Force, American Statistical Association. 2019.

Program Committee. EuroCG 2019.

Poster Committee, Grace Hopper Conference. 2015–2019.

Workshop Committee, CG Week (SoCG). 2018.

Outreach

Norm the Narwhal's Quest for Treasure, children's book. Emma Bergman, Jessi Cisewski-Kehe, and BTF. Independently Published (Amazon), October 2023.

K-12 Teacher PD events (One Week): March 2022, August 2022, Summer 2023, February 2026 (3 events).

Presentations / info sessions for K-12 teachers (30 minutes to two hours): Montana STEM Summer institute (2022), Montana Federation of Public Employee's (MFPE) Educator's conference (2024, 2025).

Middle and high school outreach events (typically about 1 hour in length and including a hands-on activity): Girls for a Change (Feb. 2020), Hawthorne Elementary STEAM Night (Spring 2020), West Yellowstone High campus visit (Spring 2022), Hardin High campus visit (Spring 2022), First Lego Competition (2022, 2023), Three Forks Elementary STEM night (March 2023), Science Olympiad Talks & Tours (2022, 2023, 2025), MSU Math Camp (2025), American Computer & Robotics Museum Summer Camp (2026), and Shadow Day Immersive Experience (2026).

Presentations / guest lectures for college students: BIOB 105 (2023, 2025), Freshman Research Symposium (2020, 2022, 2023).

Open-house style events (about three hours, where participants can walk between rooms/tables): Montana Science Center "Women in Science" Event (Feb. 2023), MSU Family Science Night (yearly event): (2020, 2022–26), and Gianforte Hall grand opening (Sept. 2026).

University and College Service

Committee Member, NACOE Distinguished Prof. Selection Committee, NACOE, 2023, 2024, 2026.

Adviser, CompTaG (research seminar and student club), MSU, 2015–Present.

Committee Member, University Teacher Education Committee (UTEC), Dec. 2021–Feb. 2024.

Center for Faculty Excellence Facutly Mentor, 2018–19 (Madison Myers), and 2020–21 (Shinjini Nandi).

Faculty Engagement Committee for NCUR 2020, May 2019–March 2020.

Organizer, Computer Science Seminar, AY 2017–2018.

Committee Member, Promoting Academic Integrity, COE, AY 2016–2017.

Department or School Service

Director of GSoC BS/MS Accelerated Program, Spring 2026–Present.

Chair, GSoC Graduate Program Requirements Revisions Committee, Spring 2026–Present.

Chair, Graduate Recruitment Committee, 2018–Present.

Committee Member, Faculty Search (TT), MSU GSoC, AY 2025–2026.

Committee Member, GSoC Retention, Tenure, and Promotion (RTP) Committee 2021, 2023, and 2025.

Director of GSoC Master's Studies, Fall 2022–Spring 2024.

Committee Member, Broadening Participation in Computing, GSoC, Fall 2020.

Adviser (and Founding Member), Upsilon Pi Epsilon (UPE) Honor Society, MSU, 2017–Present.

Committee Member, Undergraduate Curriculum, GSoC, 2015–Present.

Adviser, Association for Women in Computing (AWC), 2015–2018.

Committee Member, Faculty Search (NTT and TT), MSU GSoC, AY 2016–2017.

Committee Member, Computer Science B.A., MSU GSoC, AY 2016–2017.

Re-founding Member, Upsilon Pi Epsilon (UPE), Tulane University, 2015.

Art

CAVE: An Artscience Installation. The NeuroCave Collaborative. Holter Museum of Art, August 2017–December 2017.

Art-Math Display. The curious construction conclave, 2014. On display, Museum of Mathematics (MoMath), New York City.

News and Media

Anderson, Skip. "Montana State faculty member Brittany Fasy wins prestigious PECASE award." Montana State University, 22 Jan. 2025.

Rotlisberger, EmmaLee. "Montana State Art Student Publishes Children's Book in Collaboration with University Professors." MSU News Service, 30 Apr. 2024.

Swearingen, Marshall. "Montana State Researchers Harness Artificial Intelligence for Cancer Diagnosis." MSU News Service, 27 June 2023.

Kekatos, Mary. "How Artificial Intelligence Is Being Used to Detect, Treat Cancer – and the Potential Risks for Patients." ABC News Digital, New York, 21 July 2023. (Interviewed and quoted).

MSU "Most Valuable Professor" (MVP) Radio Broadcast, XL Country and Varsity Network, 6 November 2021.

Swearingen, Marshall. "MSU Computer Science Professor Wins Prestigious National Science Foundation Award." MSU News Service, 1 Apr. 2021.